

The Design and Physical Implementation of Seeding Robots in Deserts

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Abstract: In recent years, the land desertification has become the focus of attention in the desertified countries and areas, restricting the sustainable development of social economy. In order to solve this problem, people carry out the soil and water conservation work, using plants to prevent water and soil erosion. However, since the method of manual planting is not suitable for use in deserts, various kinds of robots are designed to sow seeds in desert areas. Inspired by tumbleweed, a new type of seeding robot in deserts is designed in this paper, which is mainly driven by wind. The robot has low energy consumption and excellent endurance. It is noteworthy that the main source of working energy for the robot is green energy such as wind and solar energy, which has no pollutant emission and is beneficial to environmental protection. In addition, the bionic structure designed for the robot makes it excellent at adapting to rugged terrain, such as sand and rocks. It can cross terrain obstacles that are difficult for the traditional desert robots to surmount. Moreover, the robot has two working shapes, which can well fulfill the requirements of marching and staying. Besides, the special design of seeding device is able to well control the quantity and time of seeding.

Key Words: Desert robots, Seeding, Deformation, Physical implementation.

1 Introduction

Land desertification is one of the important environmental issues, which expands rapidly under the dual influence of human activities and environmental climate. The United Nations Conference on Environment and Development (UNCED) presents a definition on the desertification in 1992: Desertification is the degradation of land in arid, semi-arid and semi-humid areas with drought disasters due to factors such as climate change and unreasonable economic activities of human beings. At present, the global desertification land area covers 36 million square kilometers, accounting for 1/4 of the global land area, which is equivalent to the total land of Russia, Canada, China, and the United States. In general, the desertification land area shows an increasing trend. The area is expanding at a rate of 50,000 to 70,000 kilometers per year. The desertification not only brings great harm to the ecological environment, but also severely restricts the sustainable development of social economy [1, 2]. Therefore, managing and controlling desertification as soon as possible is extremely urgent.

Generally speaking, the most effective measure to combat it is the vegetational sand-fixation. There are several issues that need to be addressed for the vegetational sand-fixation: proper seeds, suitable growing environments, and appropriate seeding methods. First of all, although the vegetation in deserts is scarce, there are still a small number of plants in depressions between the dunes, such as *Tamarix chinensis* Lour, salt-peter shrubs, and reeds. They tend to have a

quite developed root system in order to draw water from the ground to feed themselves. Therefore, it can be seen that there is no shortage of the benign environment for vegetation in deserts, but the existence of a thick layer of quicksand in deserts hinders the spread of vegetation. Beyond that, desert areas are often sparsely populated, and the transportation is inconvenient. Existing planting methods, such as artificial afforestation and aerial sowing, have difficulties and high costs.

Robots, as a kind of machines performing tasks automatically, are often used to assist or replace humans in dangerous or knotty tasks [3, 4]. It can directly receive human instructions to perform operations, or it can run pre-programmed programs. Based on this, some excellent intelligent algorithms have been developed to solve various difficult problems in the field of electronic control and robotics in recent years [5–11]. For example, in dangerous works such as high-altitude operations, resource exploration, resource development and disaster relief, it is very likely to cause injury to workers. Therefore, we can choose to use robots instead of humans to complete these tasks to avoid causing injuries. In the industrial production process, the robotic arm and the robotic hand play an important role in the assembly line operation, which reduces the consumption of human resources and the error rate [12–17]. Therefore, an automated desert robot can be designed to monitor land conditions and plant vegetation in certain strategic locations, which can help vegetation species break through the limits of environmental barriers.

However, seeding robots have defects such as poor terrain adaptability and high energy consumption, which are not suitable for use in deserts [18, 19]. For example, the traditional forms of wheeled vehicles and tracked vehicles are often used for precision seeding technology, which can effectively perform the best depth and the best distance between crops and rows on flat land for each crop type sowing. But they often do not adapt well to the sandy and gravel ter-

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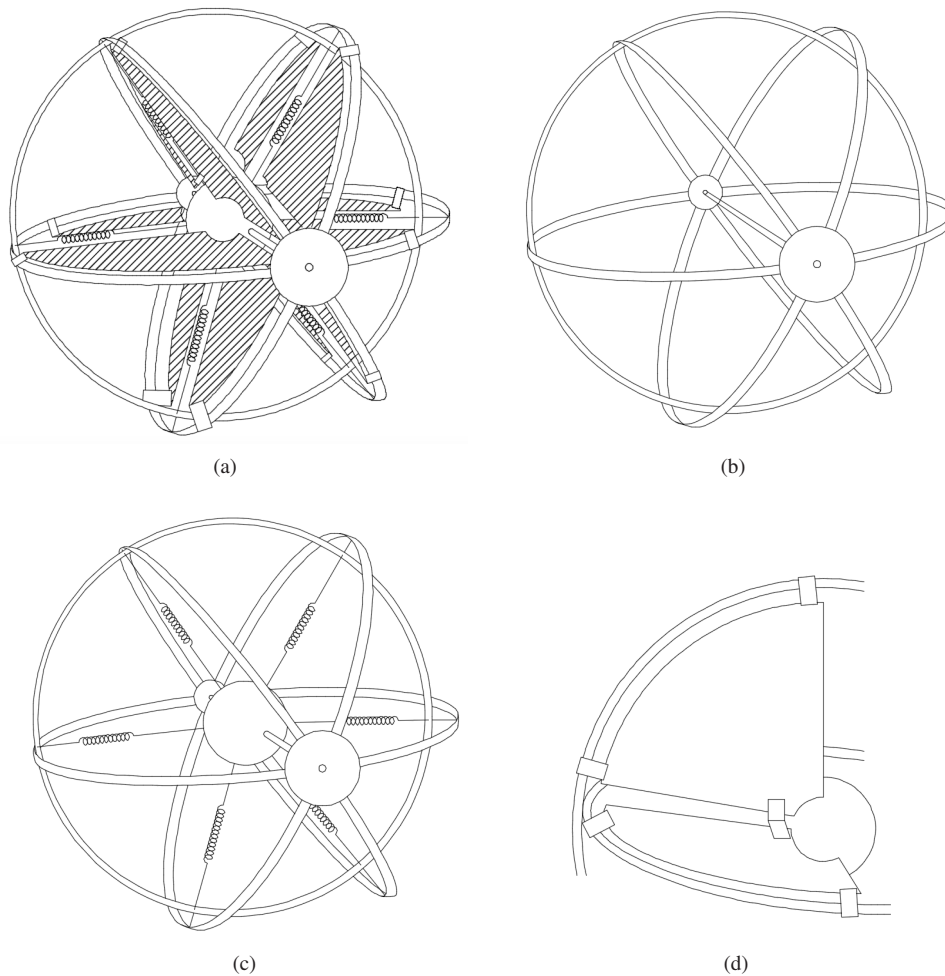


Fig. 1: External structure.

rain, which are easy to get tires into the sand and cause vehicle rollover [20–23]. Another foot-shaped robot performs a dynamic simulation of the movement of multi-legged animals. It has a high degree of freedom and can realize a variety of functions, but it is inevitable that its cost is high and the control process is very complicated [24, 25]. Additionally, most of these robots are mainly driven by batteries, thus consuming high energy to march in deserts. As a result, they all are hard to put into use on a large scale in deserts.

In order to handle the problem of planting vegetation in deserts with high efficiency and low cost, this paper proposes a novel seeding robot in deserts. Compared with the traditional seeding robots mentioned above, the seeding robot possesses superior adaptability to desert terrains. Additionally, the main source of power is the wind, which is both energy-efficient and environmentally friendly. In the following sections, a detailed introduction to the robot from the aspect of external bracing structure, internal control device and the working flow is given. Finally, conclusions are drawn and the conception of the future work is put forward.

2 Main Principle

Tumbleweed is a common plant on the Gobi [26]. When the drought comes along, it would collect roots from the soil, roll into a ball, and then roll around with the wind until the environment is suitable for growth. Then it would take root and continue to grow. As tumbleweeds roll with the wind,

they leave seeds along the way. Inspired by the tumbleweed, a new bionic robot model is proposed, which is mainly driven by wind. The robot has two working states, one is a sphere motion state, and the other is a flat-sphere stay state.

When the control center sends a movement instruction to the robot, the robot controls itself as a sphere, so that the canvases hanging on the robot's skeleton are spread out. The robot scrolls by the wind to achieve the purpose of low energy consumption and rapid movement. Apart from that, when the control center sends a stop command to the robot, the robot controls itself to become a flat-sphere and stays in place. During the work, the robot uses the mounted sensors to detect the soil quality and the surrounding environment, then transmits the detection data to the control center in real-time. In addition, the robot selects a suitable location for the autonomous seeding based on the detected data.

The seeding robot in deserts proposed in this paper is mainly composed of two parts: the external bracing structure and the internal control devices. The two parts are described in detail below.

2.1 External Bracing Structure

The main requirements of the external bracing structure design are that 1) the overall shape of the robot is suitable to march by the wind, and easy to cross various types of terrain obstacles; 2) the robot can control itself to deform

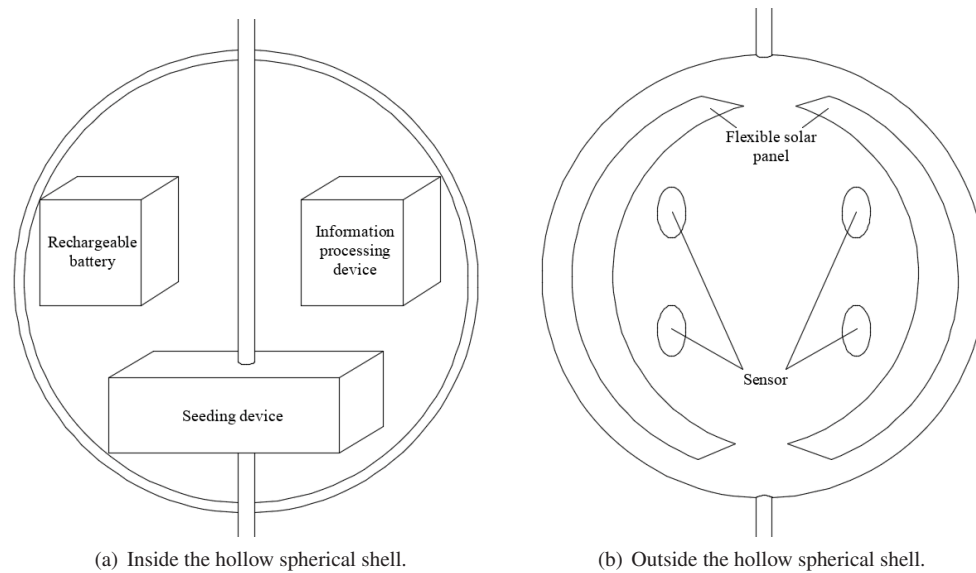


Fig. 2: Internal control device.

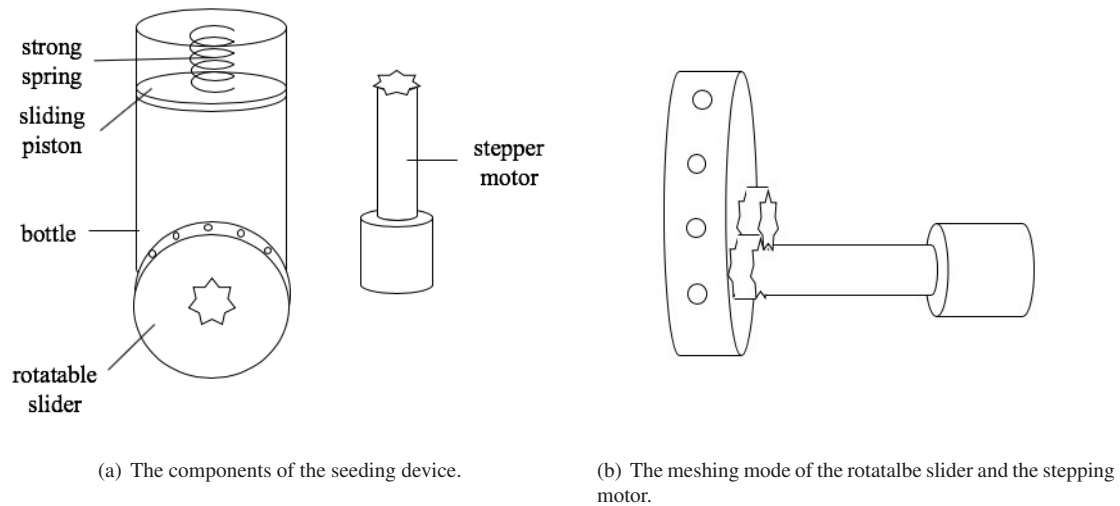


Fig. 3: Seeding device.

and change to a shape that is not easy to march by the wind to stay in place steadily; 3) the area of a robot exposed to the wind can vary with its shape; 4) the structure of the robot is balanced and stable. To meet the above requirements, the external bracing structure is designed as shown in Fig. 1(a).

The fundamental bracing form in Fig. 1(b): six elastic steel bars are used as the framework material of the bracing frame, and two ends of each steel bar are respectively connected with two circular bases to form a sphere. Furthermore, a bracing ring is placed in the middle position inside the bracing frame to ensure the stability of the sphere.

As shown in Fig. 1(c), the installation platform for installing the internal control devices is a hollow spherical shell, which is placed in the center of the robot and connected to the bracing frame by springs through the bracing ring. Moreover, the expansion link passes through the hollow spherical shell of the installation platform, and its two ends are respectively connected to two bases. It is used to change the shape of the robot.

As shown in Fig. 1(d), the fan-shaped canvases are con-

nected to the external bracing frame and the bracing ring through a sliding connection device. Adjacent canvases are connected near the hollow spherical shell to make them fully unfolded, thereby increasing the force area of wind force and pushing the robot to march.

2.2 Internal Control Device

The basic mode of the robot is set up in the above-mentioned external structure. The following is an introduction to the control device mounted on the robot.

As shown in Fig. 2(a), inside the hollow spherical shell, the devices carried by the robot include rechargeable battery, information processing device, and seeding device. As shown in Fig. 2(b), on the surface of the hollow spherical shell, flexible solar panels and various sensors (light sensor, temperature sensor, humidity sensor, oxygen sensor, carbon dioxide sensor, etc.) are placed.

Rechargeable battery provides a source of power for the robot's working processes of control, detection, and seeding, ensuring the normal operation of information processing de-

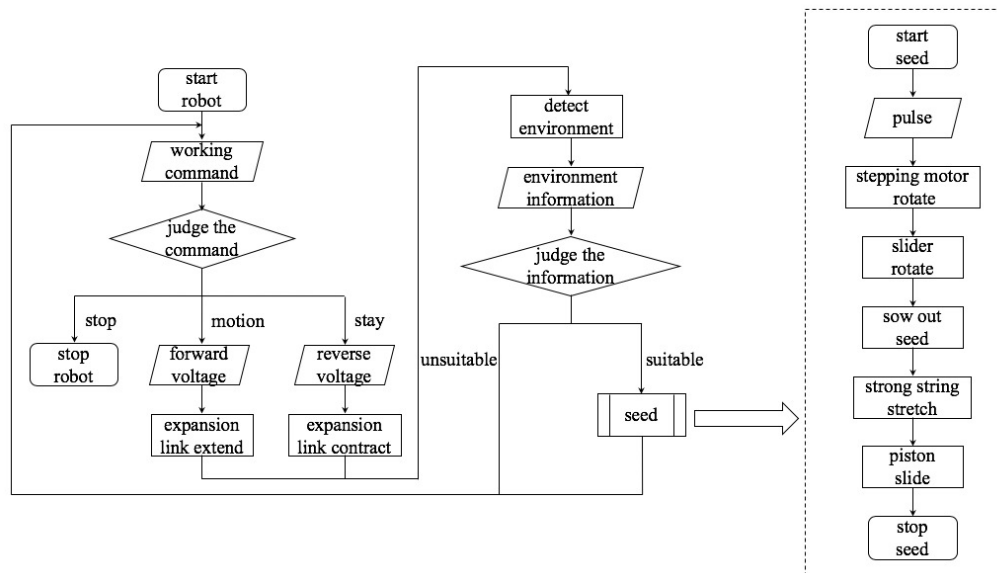


Fig. 4: Working process.

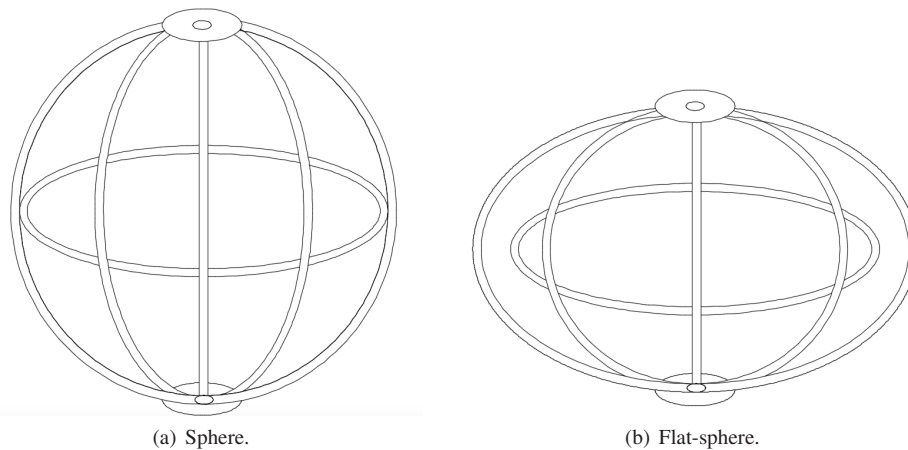


Fig. 5: Two working states.

vice, seeding device, and sensors. The flexible solar panels are exposed to sunlight, which convert sunlight into electricity and charge the battery to enhance the endurance of the robot. Moreover, various kinds of sensors collect environmental information and transmit it back to the control center in real-time through the information processing device so as to find a suitable environment for seeding.

2.2.1 Information Processing Device

The information processing device is provided with a single-chip microcomputer, a transmitting device, a receiving device, an electrical quantities detection device, and a positioning device to maintain the normal work of the robot.

2.2.2 Seeding Device

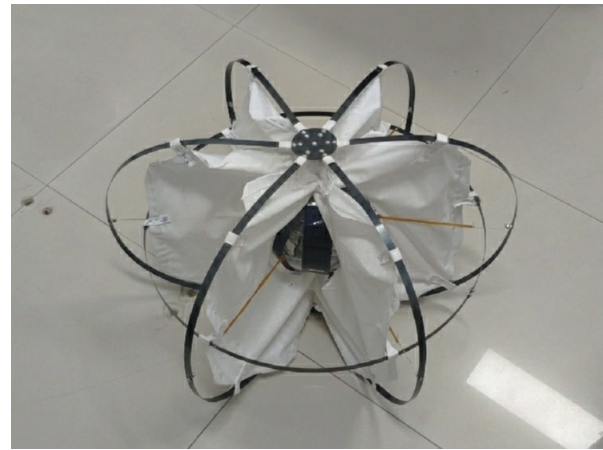
As shown in Fig. 3(a), the seeding device includes a bottle, a sliding piston, a strong spring, a rotatable slider, and a stepper motor. Inside the bottle, a sliding piston with the same size as the bottom of the bottle is placed. The two ends of the strong spring are connected to the sliding piston and

the bottom of the bottle. Beyond that, a rotatable slider is installed at the mouth of the bottle. The rotatable slider is cylindrical with the gear-shaped protrusions on the bottom surface to mesh with the gear of the stepping motor, and the corresponding meshing mode is shown in Fig. 3(b). Equivalent depressions are set on the side of the cylindrical slider. The depressions are the same size as the seeds to be sown. Besides, the distance between each depression is set according to the rotation angle of the stepper motor. After putting the seeds into the bottle, the piston is pressed to the bottom of the bottle, and the spring is compressed. Then, install the rotatable slider on the mouth of the bottle, stably meshing the gears of both the slider and the stepper motor.

When the robot starts the seeding function, the stepping motor controlled by the pulse rotates a fixed angle, and the slider rotates accordingly, sowing out the seeds embedded in the depression on the side of the slider, thereby completing the task of seeding. After the seeds are sowed, the space taken up by seeds between the piston and the slider becomes smaller. The strong spring attached to the piston pushes it to slide, forcing the air between the piston and the slider out, to ensure that there always has a seed in every depression on



(a) Sphere.



(b) Flat-sphere.

Fig. 6: Actual pictures.

the side of the slider which can be sowed.

3 Working Process

The seeding robot in deserts has two working states, one is a sphere motion state, the other is a flat-sphere staying state. The workflow of the robot is shown in Fig. 4.

When the robot is successfully launched, the remote control center sends a working command to it. Then, the receiving device mounted on the robot receives the command, and the robot starts the sphere motion state. In this sense, the single-chip microcomputer outputs a forward voltage to control the expansion link to extend, so that the robot as a whole becomes a sphere, which is shown in Fig. 5(a). There is no deformation of the spring that fixes the installation platform to the bracing frame, and the fan-shaped canvases connected to the robot is fully unfolded. The area that the robot receives wind power is greatly increased. Further, the robot marches in the wind.

When the remote control center sends a staying command to the robot, the receiving device mounted on the robot receives the command, and then the robot starts flat-sphere staying state. In this sense, the single-chip microcomputer outputs the reverse voltage to controls the expansion link to contract, so that the overall shape of the robot becomes a flat-sphere, which is shown in Fig. 5(b). The spring that fixes the installation platform to the bracing frame is stretched, and the fan-shaped canvas connected to the robot is no longer unfolded. The area that the robot receives wind power is greatly reduced. Also, the flat-sphere is not easy to roll, thus the robot can stay in place stably.

During the work time of the robot, the sensors mounted on the outside of the installation platform are powered on and detect the surrounding environment. The information processing device transmits the environmental information detected by sensors and the position information detected by the positioning device to the control center for archiving through the transmitting device. This would facilitate people to carry out the desertification control. When the environmental information detected by sensors is suitable for seeding, the seeding device would be activated. Besides, the single-chip microcomputer provides a pulse signal for the stepping motor. The motor rotates a fixed angle and drives

the rotatable slider in the seeding device to rotate to broadcast the seed.

In particular, when the electrical quantities detection device detects that the battery level is less than 5%, the single-chip microcomputer would automatically control the robot to change into a flat-sphere and transmits the location information to the control center. The work does not resume until the flexible solar panel recharges the battery fully. For illustration, the actual pictures of the seeding robot in deserts are shown in Fig. 6. More specifically, Fig. 6(a) is the sphere motion state, and Fig. 6(b) is the flat-sphere staying state.

4 Future Work

The current model of the robot can control the timing and amount of seeding by controlling the rotation of the stepper motor. In the future, we will further explore and expand the seeding function. The robot's mimicry object—tumbleweed—is an effective seed propagator. Its spherical shape allows it to roll and bounce all over the ground, and each bounce leaves seeds along the way. The terrain where it is easy to bounce and drop seeds is mostly ditch and concave, which happens to be quite suitable for growth. Using this feature, we will design a new seeding device that automatically broadcasts seeds when the robot has a large bounce, so as to reduce the burden on the robot and make it simpler and lighter.

In addition, we will also carry out a transformation of the installation platform and the expansion link. Further, we will build a planetary gear system inside the installation platform to replace the expansion link to create torque necessary to change the shape of the robot.

Thirdly, we will install a kinetic energy generator inside the installation platform so that the robot can obtain more energy to feed its motors and computer systems.

5 Conclusions

In this paper, a new seeding robot in deserts has been proposed, which uses tumbleweed as a biomimetic object and relies on the wind to roll forward in the working state. The sensors mounted on it detect the surrounding environment information to help it find a suitable location to sow the seeds. Flexible solar panels convert light energy into elec-

tricity, providing energy for the robot control and inspection work. Compared with the traditional seeding robot in deserts, this robot is simpler to operate and is mainly driven by wind, thereby greatly reducing energy consumption. Besides, it can switch between two different shapes for different purposes. The shape of sphere allows the robot to cross more complex terrain obstacles, and the shape of flat-sphere allows it to stay in the expected position more stably.

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